Ruyi Pan

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Research Interests

High-dimensional statistical inference; multivariate and multi-view data analysis; dependence modeling; statistical methods for biomedical and neuroimaging applications.

Education

University of Toronto — Sept. 2022 – June 2027 (Expected)
PhD Candidate in Statistical Sciences
GPA: 4.00/4.00
Supervisors: Jun Young Park, Nancy Reid, and Aristotle N. Voineskos

University of Toronto — Sept. 2020 – June 2021
MSc in Statistics

University of Waterloo — Jan. 2018 – Aug. 2020
Bachelor of Computer Science, Honours; Minor in Statistics

Working Papers and Research in Progress

[1] **Pan, R.**, He, Y., & Park, J. Y. (2026). Boosting multi-view association testing via devariation. arXiv:2603.26981. [*Invited resubmission, JASA – Theory and Methods*].

[2] **Pan, R.**, Nieto-Barajas, L. E., & Craiu, R. V. Multivariate temporal dependence via mixtures of rotated copulas. arXiv:2403.12789. [*Revision under Canadian Journal of Statistics*].

[3] Chaposhloo, M., Oliver, L. D., Power, J. D., Byambadorj, J., Yu, J.-C., **Pan, R.**, Bingham, K. S., Agarwal, M., Hahn, M., Foussias, G., Mulsant, B. H., Flint, A. J., Voineskos, A. N., Hawco, C., & Neufeld, N. H. Precision functional mapping in individuals with schizophrenia spectrum disorders and psychotic depression: feasibility and connectome fingerprinting. [*Manuscript in preparation*].

[4] Bai, R., Ng, W. A. Y., **Pan, R.**, & Park, J. Y. A score-based test for heritability of intermodal coupling in neuroimaging. *In progress.*

[5] **Pan, R.**, Oliver, L. D., et al. Brain connectivity foundation model: a knowledge-grounded AI system for human connectome discovery. *In progress.*

Publications

[1] Chen, Y., Rajji, T. K., Zhukovsky, P., Abdulrasul, H., Fischer, C. E., Flint, A. J., Herrmann, N., Ho, N. C. W., Lanctot, K., Lee, A., Ma, C., Mah, L., Mansour, S., **Pan, R.**, Park, J. Y., Pollock, B. G., Mulsant, B. H., & Voineskos, A. N. (2026). Impact of cognitive remediation combined with transcranial direct current stimulation on gray matter macrostructure and microstructure in older adults at risk for Alzheimer's dementia. *Biological Psychiatry: Global Open Science*. DOI: 10.1016/j.bpsgos.2026.100834.

[2] **Pan, R.**, Weinstein, S. M., Tu, D., Hu, F., Tanrıverdi, B., Zhang, R., Vandekar, S. N., Baller, E. B., Gur, R. C., Gur, R. E., Alexander-Bloch, A. F., Satterthwaite, T. D., & Park, J. Y. (2025). Mapping individual differences in intermodal coupling in neurodevelopment. *Imaging Neuroscience*. 3: IMAG.a.156. DOI: 10.1162/IMAG.a.156.

[3] Perreault, S., Tang, Y., **Pan, R.**, & Reid, N. (2025). Inference for overparametrized hierarchical Archimedean copulas. *Journal of Multivariate Analysis*. 210: 105483. DOI: 10.1016/j.jmva.2025.105483.

[4] **Pan, R.**, Nieto-Barajas, L. E., & Craiu, R. V. (2025). Bayesian nonparametric mixtures of Archimedean copulas. *Journal of Agricultural, Biological and Environmental Statistics*. DOI: 10.1007/s13253-025-00703-8.

[5] Rashidi, A. G., Oliver, L. D., Moxon-Emre, I., Hawco, C., Dickie, E. W., **Pan, R.**, Secara, M. T., Yu, J.-C., Szatmari, P., Desarkar, P., Foussias, G., Buchanan, R. W., Malhotra, A. K., Lai, M.-C., Voineskos, A. N., & Ameis, S. H. (2026). Comparative analysis of social cognitive and neurocognitive performance across autism and schizophrenia spectrum disorders. *Schizophrenia Bulletin*. 52(1): sbaf005. DOI: 10.1093/schbul/sbaf005.

[6] **Pan, R.**, Dickie, E. W., Hawco, C., Reid, N., Voineskos, A. N., & Park, J. Y. (2024). Spatial-extent inference for testing variance components in reliability and heritability studies. *Imaging Neuroscience*. 2: imag-2-00058. DOI: 10.1162/imag_a_00058.

[7] **Pan, R.**, Diaz-Martinez, J., Gronsbell, J., & Touma, Z. (2022). Screening for cognitive impairment with the Automated Neuropsychological Assessment Metrics in patients with systemic lupus erythematosus. *Journal of Rheumatology*. 49(7): 840–841.

Awards and Honours

Donald A.S. Fraser Doctoral Award, Department of Statistical Sciences, University of Toronto 2026
Awarded for PhD students for significant contributions to statistical theory.

Student Paper Award (Runner-Up), ASA Statistics in Imaging Section 2025

Travel Grant, Institute for Mathematical and Statistical Innovation Workshop, University of Chicago 2024

Early Research Excellence Award, Department of Statistical Sciences, University of Toronto 2024
Awarded for junior PhD students for research excellence.

Faculty of Arts and Science Recognition of Excellence Award, University of Toronto 2022
Awarded for incoming PhD students who demonstrate academic excellence.

CANSSI Doctoral Recruitment Award, University of Toronto 2022

Andrews Academic Achievement Award, University of Toronto 2021
Awarded for outstanding academic work in the Master's program.

Douglas Lankshear Scholarship for Women in Computer Science, University of Waterloo 2020
Awarded for academic excellence in undergraduate computer science.

Conference Presentations

Talks

Devariation: a robust approach to improve statistical power in high-dimensional multi-view association testing. *Joint Statistical Meetings (JSM)* 2025

Improving statistical power of multi-modal associations via de-variation. *Eastern North American Region (ENAR) Spring Meeting* 2025

Improving statistical power of multi-modal associations via de-variation. *CMStatistics* 2024

A fast and powerful spatial-extent inference method for reliability of task-fMRI. *Joint Statistical Meetings (JSM)* 2023

Posters

Improving statistical power of multi-modal associations test via de-variation. *Institute for Mathematical and Statistical Innovation (IMSI) Workshop, University of Chicago* 2024

Fast and powerful spatial-extent inference for testing variance components in reliability and heritability studies. *Eastern North American Region (ENAR) Spring Meeting* 2024

Fast and powerful spatial-extent inference for testing variance components in reliability and heritability studies. *Statistical Methods in Imaging (SMI)* 2023

Screening for Cognitive Impairment with the Automated Neuropsychological Assessment Metrics in Patients with Systemic Lupus Erythematosus. *ACR Convergence* 2021

Academic Service

Member, CANSSI Ontario Student PhD and Research Network (SPARK) Program 2025–2026

Organizing Committee, The Fast and Curious II: MCMC in Action, University of Toronto 2025

Session Chair, University of Toronto Statistics Graduate Student Research Day 2024, 2025

Session Chair, ENAR Spring Meeting 2024

Reviewer, *Biometrics*; *Computational Statistics & Data Analysis*; *Annals of Applied Statistics* (under guidance of Prof. Jun Young Park)

Volunteer, Statistical Sciences Graduate School Information Session, University of Toronto 2024

Teaching Experience

University of Toronto, Teaching Assistant 2021 – Present

STA437H1 – *Methods for Multivariate Data* *Winter 2026*
STA2101 – *Methods of Applied Statistics, Graduate* *Fall 2024, 2025*
STA442H1 – *Methods of Applied Statistics, Undergraduate* *Fall 2024, 2025*
STA347H1 – *Probability* *Summer 2025*
STA238H1 – *Probability, Statistics and Data Analysis II* *Winter 2024, 2025*
STA237H1 – *Probability, Statistics and Data Analysis I* *Fall 2022, 2023, 2026*
STA221H1 – *The Practice of Statistics II* *Winter 2026*
STAB57H3 – *An Introduction to Statistics* *Winter 2021*
STAB22H3 – *Statistics I* *Winter 2021*
CSCA08H3 – *Introduction to Computer Science I* *Winter 2021*

Research and Professional Experience

Centre for Addiction and Mental Health (CAMH) – *Research Analyst* Sept. 2022 – Present
Statistical and neuroimaging analysis for interdisciplinary mental health research.

Goldman Sachs – *Quantitative Strategist Summer Associate, Multi-Asset Solutions* June 2026 – Aug. 2026
Quantitative methods and AI-based tools for institutional portfolio management.

University of Toronto – *Research Assistant* Apr. 2021 – Aug. 2021
Statistical analysis of electronic health records and statistical communication research.